Status	Finished
Started	Sunday, 18 January 2026, 8:09 PM
Completed	Sunday, 18 January 2026, 9:19 PM
Duration	1 hour 9 mins
Marks	17.50/20.00
Grade	**8.75** out of 10.00 (**87.5**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Consider the following time series containing the yearly publications of a research group:

Year	Books	Conference papers	Journal papers
2020	1	3	1
2021	1	2	2
2022	0	4	1
2023	1	2	1

Which of the following statements is true about this time series?

- [ ] This time series is in the long format
- [x] This time series is in the wide format.

Your answer is correct.

We have a multivariate time series, since we have 3 different time series: number of books published / year; number of conference papers published / year and number of journal papers published / year.

When we have a wide format, each time series is a column, just like in our case.

The correct answer is: This time series is in the wide format.

Question **2**

Correct

Mark 1.00 out of 1.00

We use the drift model to compute forecasts and the corresponding prediction intervals for three consecutive time steps. The model returned the prediction intervals from below. Which is the prediction interval for the first future time step?

- [ ] [1.07 - 4.09]
- [ ] [0.74 - 4.43]
- [ ] It cannot be determined.
- [x] [1.52 - 3.63]

Your answer is correct.

As we progress into the future, the prediction intervals become wider. So for the first time step, we need the narrowest interval.

The correct answer is: [1.52 - 3.63]

Question **3**

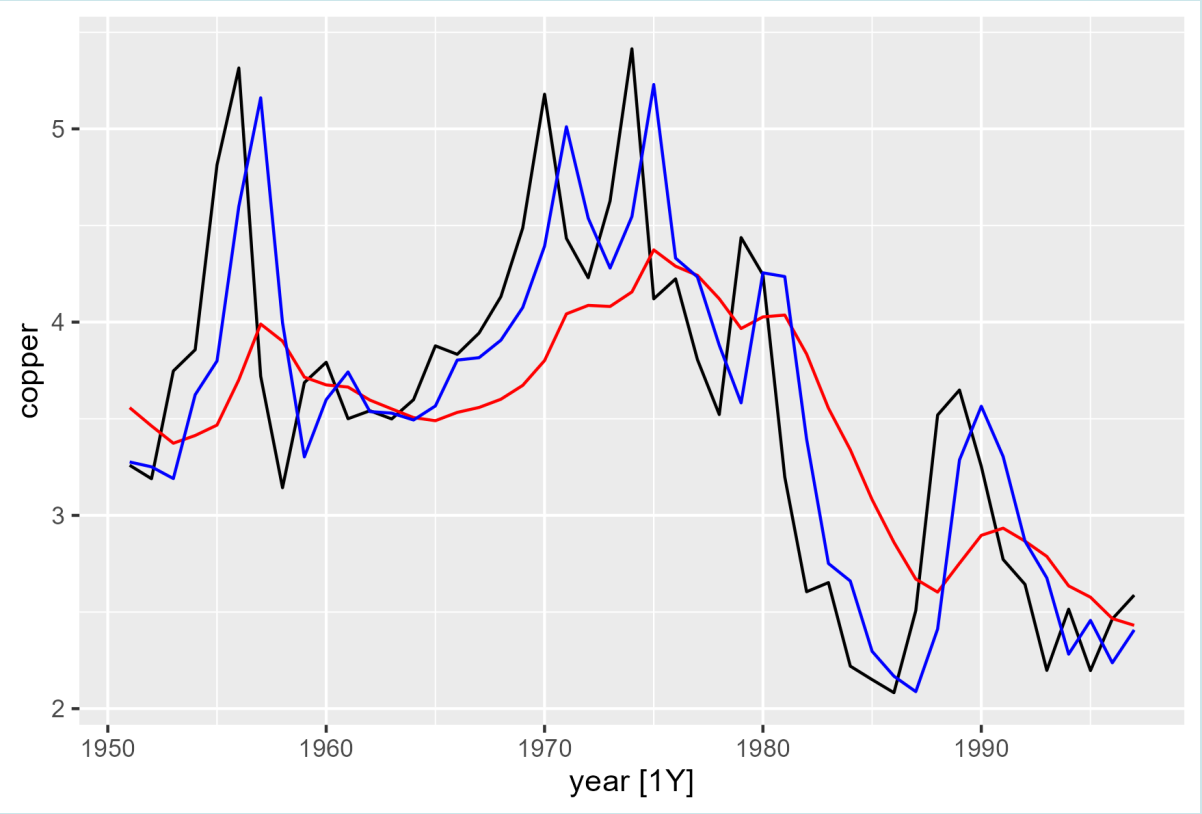
Correct

Mark 1.00 out of 1.00

On a time series about copper sales, I have fitted two ETS models, both have additive error, additive trend and no season. For one of them the value of alpha was set to 0.2, for the other, it was set to 0.8.

Below, you can see the plot of the original data (black line), and the fitted values for the two models.

Which of them is for the model with alpha = 0.2?



- [x] The red one ✓
- [ ] The blue one
- [ ] It cannot be determined, it depends on the data.

Your answer is correct.

Alpha determines how important recent observations are, since it is the coefficient for the previous observation. High values of alpha mean that the model considers the previous

observation in a significant manner, so the fitted values will follow the shape of the original data (blue line).

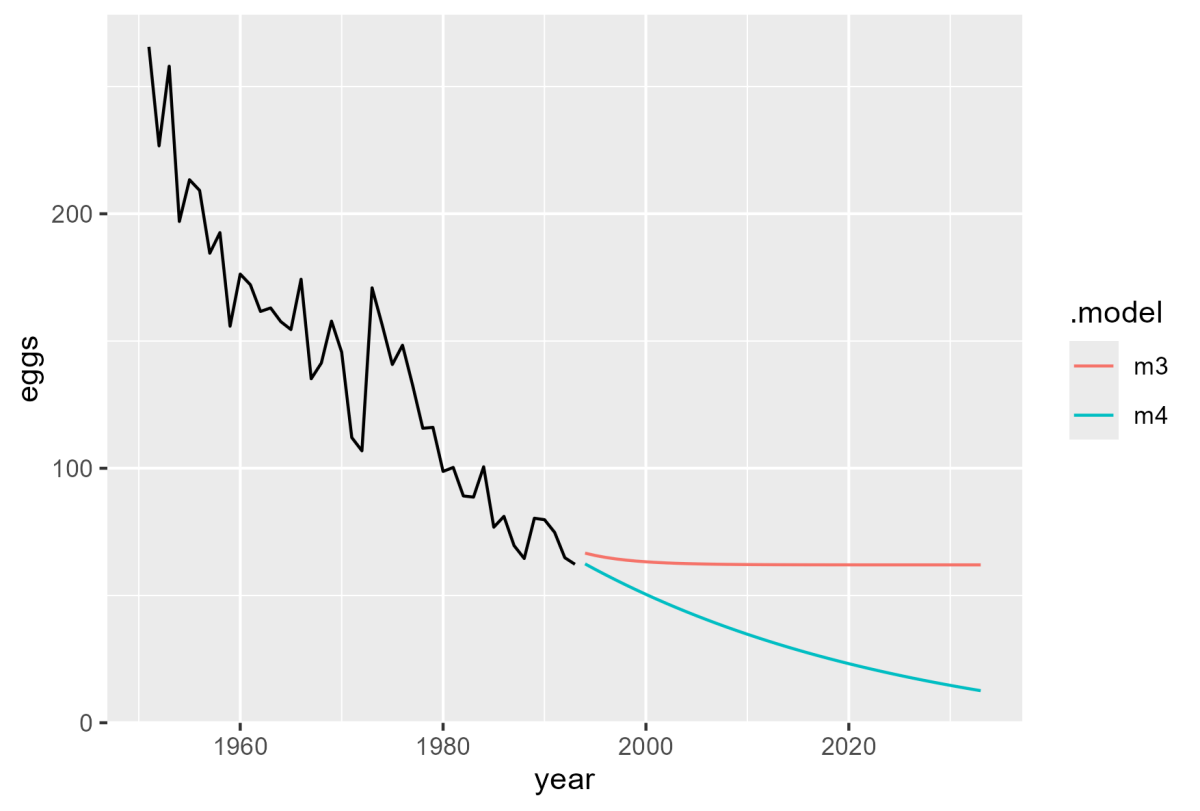
The correct answer is: The red one

Question **4**

Correct

Mark 1.00 out of 1.00

On a time series about egg prices, I have fit two ETS models: they both have additive error, no seasonality and damped trend. One model has a value 0.8 for phi, while the other has 0.97. Below, you can see the original data and the forecast of the two models for the next 40 years. Which one is the model with phi = 0.97?



- [ ] The red one.
- [x] The blue one.
- [ ] It cannot be determined, it depends on the data.

Your answer is correct.

Phi determines how fast the forecast line becomes flat. Phi = 1, means no dumping => the forecast never becomes flat. So the blue line is for phi = 0.97 and the red is 0.8.

The correct answer is: The blue one.

Question **5**

Correct

Mark 1.00 out of 1.00

Assume that you have 41 observations of quarterly data and you want to build a baseline model on it. Then you build the same baseline model only on the last 39 observations of the data. For which baseline could we have the biggest difference of the forecasts produced by the two models?

- [ ] Seasonal naive method
- [ ] Naive method
- [x] Drift method
- [ ] Mean method

Your answer is correct.

41 observations of quarterly data mean exactly 10 years plus one quarter, which means that the first and last observation are for the same quarter (ex. Q1).

Mean method - mean of the observations. For the first model we have mean of 40 observations, for the second, mean of 38 observations. The difference is probably not that big.

Naive method - just repeat the last observation. Since the second model is build on the last 38 observations, the last observation is the same for both, so they actually forecast exactly the same thing.

Seasonal naive method - just repeat the last season. This is going to be exactly the same.

Drift method - joins with a line the first and last observation and continues the line to forecasts. For 41 observations, you join data from the same quarter, so the model captures the true trend in data. For 39 observations, data is from different quarters (ex. Q1 and Q3) which means that you probably joined a "high" and a "low" value and will get a line with a steep slope and will get very different forecasts.

The correct answer is: Drift method

Question **6**

Correct

Mark 1.00 out of 1.00

Assume that we have a time series with 40 observations. If we build a mean model on it, how many fitted values will we have?

- [ ] 39
- [ ] 38
- [ ] 36
- [ ] It depends on the value of the parameter we pass to the function which produces the fitted values.
- [x] 40
- [ ] It cannot be determined.

Your answer is correct.

Mean model computes the mean value of the observations and uses this as fitted value. Will will have (the same) fitted value for all 40 observations.

The correct answer is: 40

Question **7**

Correct

Mark 1.00 out of 1.00

Assume that we have a time series with 40 observations. If we build a naive model on it, how many fitted values will we have?

- [ ] 38
- [ ] It cannot be determined.
- [ ] 40
- [ ] It depends on the value of the parameter we pass to the function which produces the fitted values.
- [ ] 36
- [x] 39

Your answer is correct.

Naive model repeats the last value as fitted value. For the first observation we will not have a fitted value, because there is no previous value to be used.

The correct answer is: 39

Question **8**

Incorrect

Mark 0.00 out of 1.00

Assume that we have a time series with 40 observations. If we build a seasonal naive model on it, how many fitted values will we have?

- [ ] 39
- [ ] 40
- [ ] 36
- [x] It depends on the value of the parameter we pass to the function which produces the fitted values. ⊗
- [ ] 38
- [ ] It cannot be determined.

Your answer is incorrect.

Seasonal naive model repeats the last season as fitted value (more precisely, for every observation it repeats the corresponding value from the previous season (for ex. Q1). So we will not have fitted values for the first season. But since the question did not specify what kind of data we have, we cannot determine how many fitted values will be missing. For quarterly data, it would be 4 missing values. For monthly, it would be 12.

The correct answer is: It cannot be determined.

Question **9**

Correct

Mark 1.00 out of 1.00

How many nodes are in a Neuran network autoregressive model: NNAR(5, 2)?

- [x] 8
- [ ] 10
- [ ] 5
- [ ] 7
- [ ] 2

Your answer is correct.

The first value, 5, denotes the number of nodes in the input layer. The second value, 2, denotes the number of nodes on the hidden layer. And we implicitly have one node on the output layer.

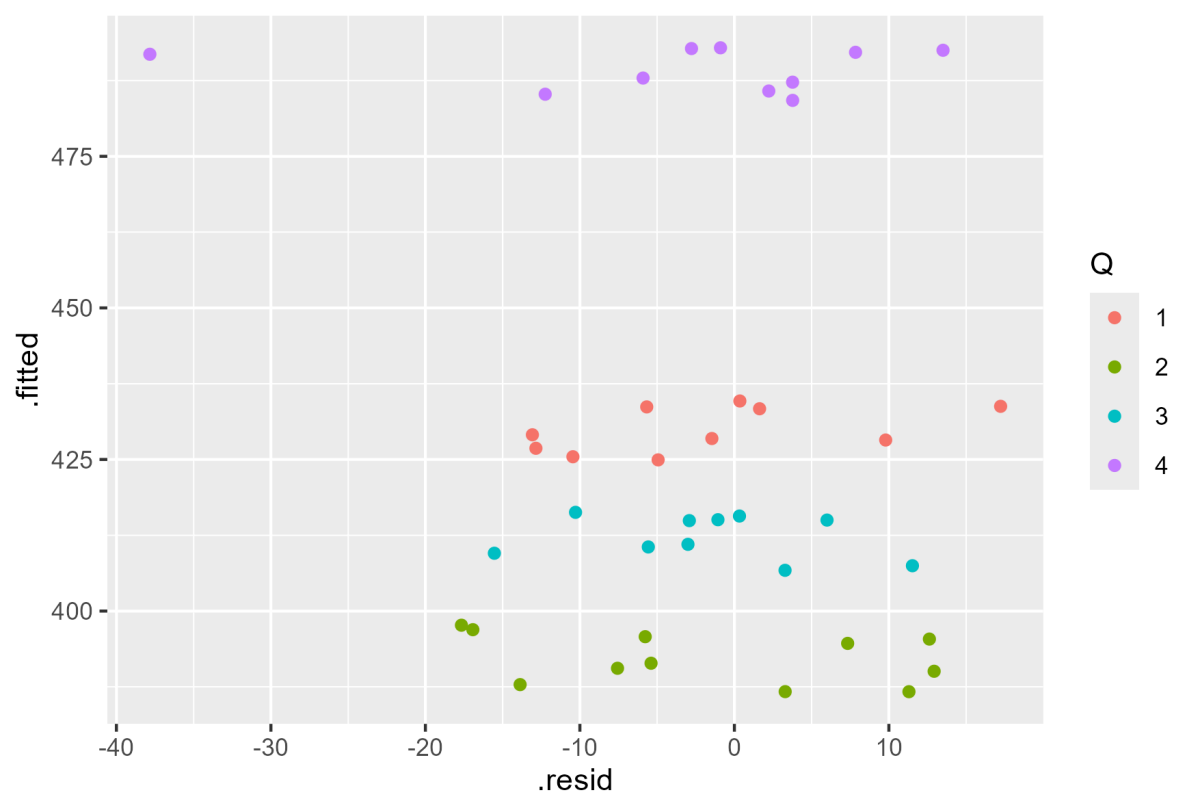
The correct answer is: 8

Question **10**

Correct

Mark 1.00 out of 1.00

After training an ETS model on a quarterly time series, we built the following plot of the fitted values versus the residuals. Which of the statements below are true?



- [ ] The highest error is for an observation from Q1.
- [ ] For a good model, the points should be around the main diagonal.
- [ ] The errors for data from the 4th quarter are in general higher than the error for the data from the other quarters.
- [x] The model seems to be a good one.

Your answer is correct.

The y axis contains the fitted values, and the x axis contains the residuals.

The fact that the purple dots are above the others, does not mean that their residual is higher, it simply means that they have higher fitted values (probably because these are the high values of the seasonal period). Their residuals are the described by how scattered they are according to the

x axis, and they are not more scattered than the residuals for the other quarters (with the exception of the outlier in the upper left corner, but the answer mentioned "in general").

For the highest error we need to look at the leftmost and rightmost point. Leftmost if the purple one in the upper left corner with a residual of about -38, while the rightmost point is one from Q1 with a value of about 17. -38 is a bigger error than 17, since it is further away from the ideal 0 residual.

When we look at a scatter plot of residuals vs. fitted values, for a good model we see no pattern in it, and this is true here.

A scatterplot centered around the main diagonal would mean that observations with a higher fitted value have higher residuals as well, which is not something we want.

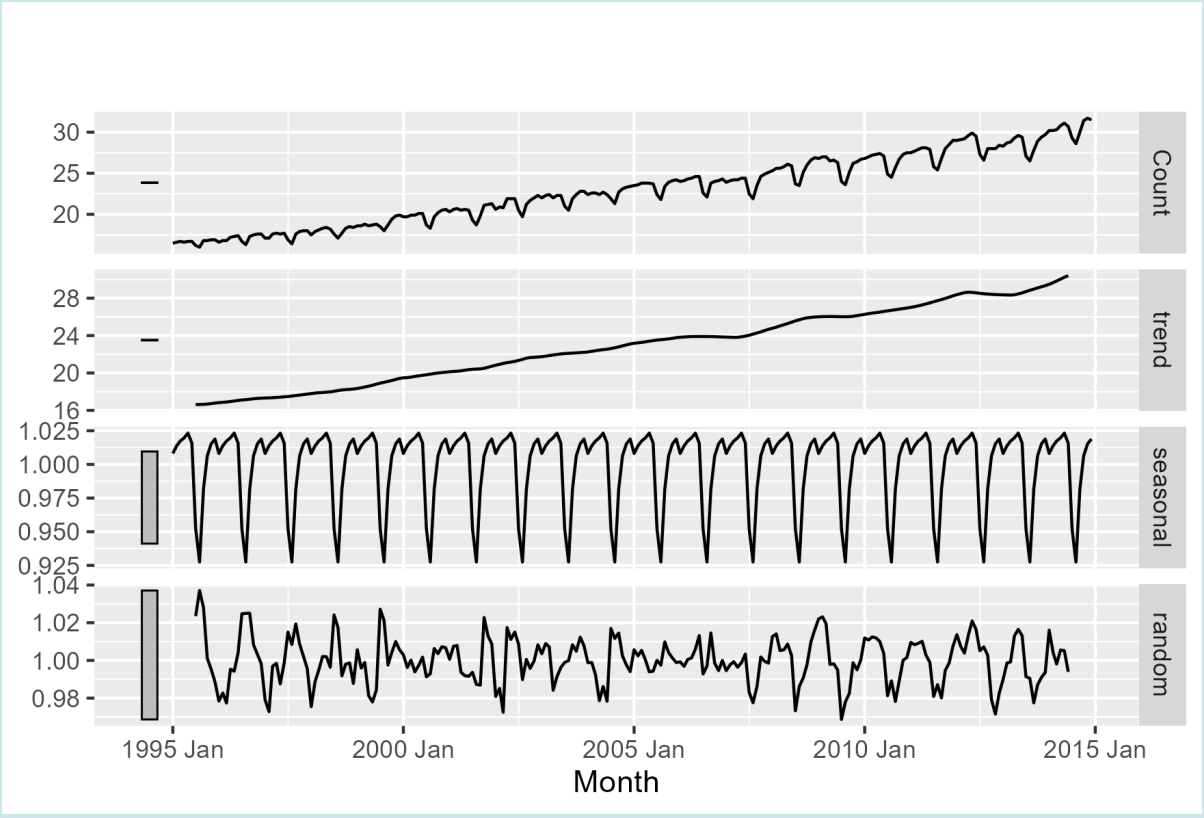
The correct answer is: The model seems to be a good one.

Question **11**

Correct

Mark 1.00 out of 1.00

Consider the following decomposition of a time series. Which of the following statements are true?



- [x] It is multiplicative decomposition ✓
- [ ] It is additive decomposition.
- [ ] The data has several seasonal patterns.
- [x] It is classical decomposition. ✓
- [ ] It is STL decomposition.

Your answer is correct.

If we look at the panel with the seasonal components, we can see that it has values between 1.025 and 0.925, so it is approximately, around the value 1. This is the characteristic of a multiplicative decomposition.

STL cannot do multiplicative decomposition, so it is certainly not STL. If we look at the trend component, we can see missing values at the beginning and end of the plot. This is due to the MA

not being able to estimate the first and last few values, and this is characteristic for classical decomposition.

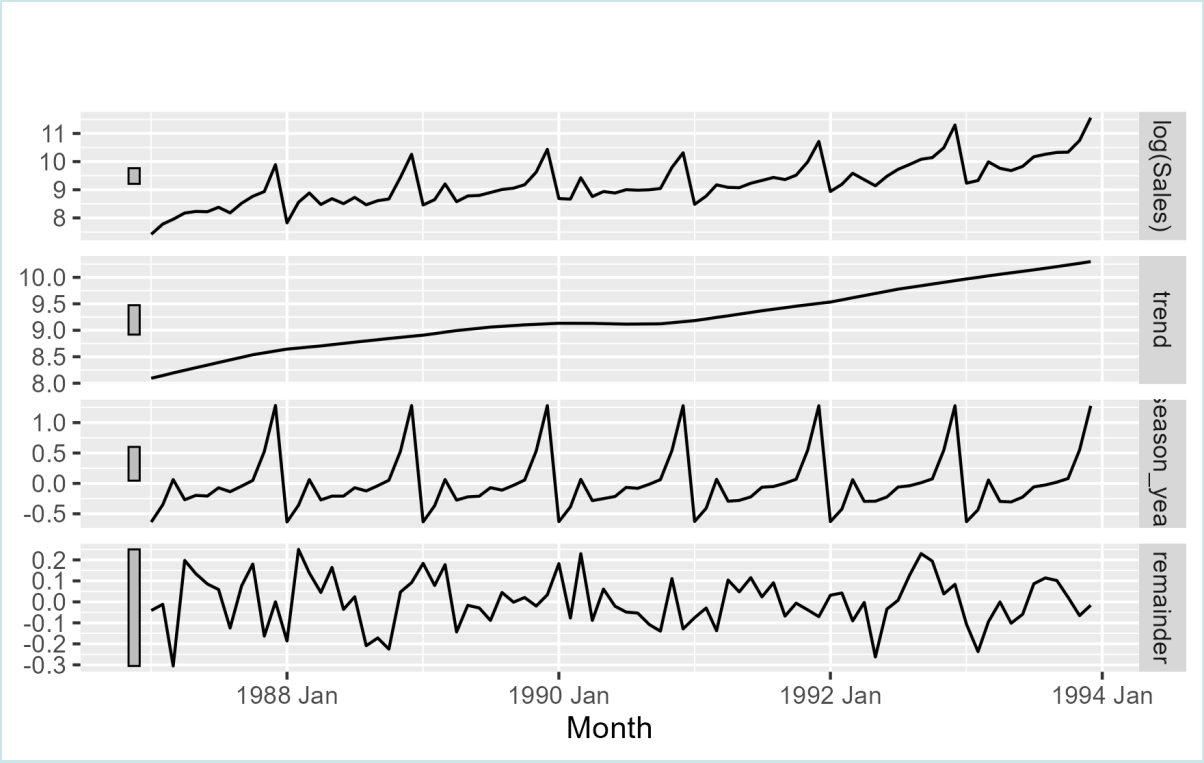
The correct answers are: It is multiplicative decomposition, It is classical decomposition.

Question **12**

Correct

Mark 1.00 out of 1.00

Consider the following decomposition of a time series. Which of the following statements are true?



- [x] It could be STL decomposition.
- [x] It is additive decomposition.
- [ ] The trend in the data is not significant, since the grey bar in front of the trend panel is quite small.
- [ ] It is multiplicative decomposition.
- [ ] It is classical decomposition.

Your answer is correct.

Since the seasonal components is centered around 0, this is additive decomposition.

Since all observations have an estimate for the trend, this cannot be classical decomposition, it could be STL (it actually is).

The trend component is actually strong (ranging from 8 to 10), whereas the error (with a long grey bar) has a range from -0.3 to 0.2.

The correct answers are: It is additive decomposition., It could be STL decomposition.

Question **13**

Correct

Mark 1.00 out of 1.00

The STL decomposition of a time series returned the following components.

What is the seasonally adjusted value for the first observation (1995 Jan)?

```
A dable: 120 x 7 [1M]
Key:     .model [1]
:        TC = trend + season_year + remainder
  .model     Month    TC  trend season_year remainder
  <chr>      <mth> <dbl>  <dbl>       <dbl>     <dbl>
1 STL(TC) 1995 Jan 43882 41146.        2890.    -155.
2 STL(TC) 1995 Feb 39912 41678.       -2399.     633.
3 STL(TC) 1995 Mar 44341 42210.        2213.     -81.7
4 STL(TC) 1995 Apr 40736 42742.       -2423.     417.
5 STL(TC) 1995 May 42142 43274.        -199.    -933.
6 STL(TC) 1995 Jun 41575 43807.       -2198.     -34.2
7 STL(TC) 1995 Jul 42948 44339.       -1006.    -385.
8 STL(TC) 1995 Aug 45053 44868.         300.    -115.
9 STL(TC) 1995 Sep 45386 45398.        -437.     426.
0 STL(TC) 1995 Oct 48851 45927.        2345.     579.
```

Answer: 40992

The seasonally adjusted value is the actual value, minus the seasonal component (or the trend component plus the remainder). Here the two values differ by 1, since the components are not fully displayed.

The correct answer is: 40992

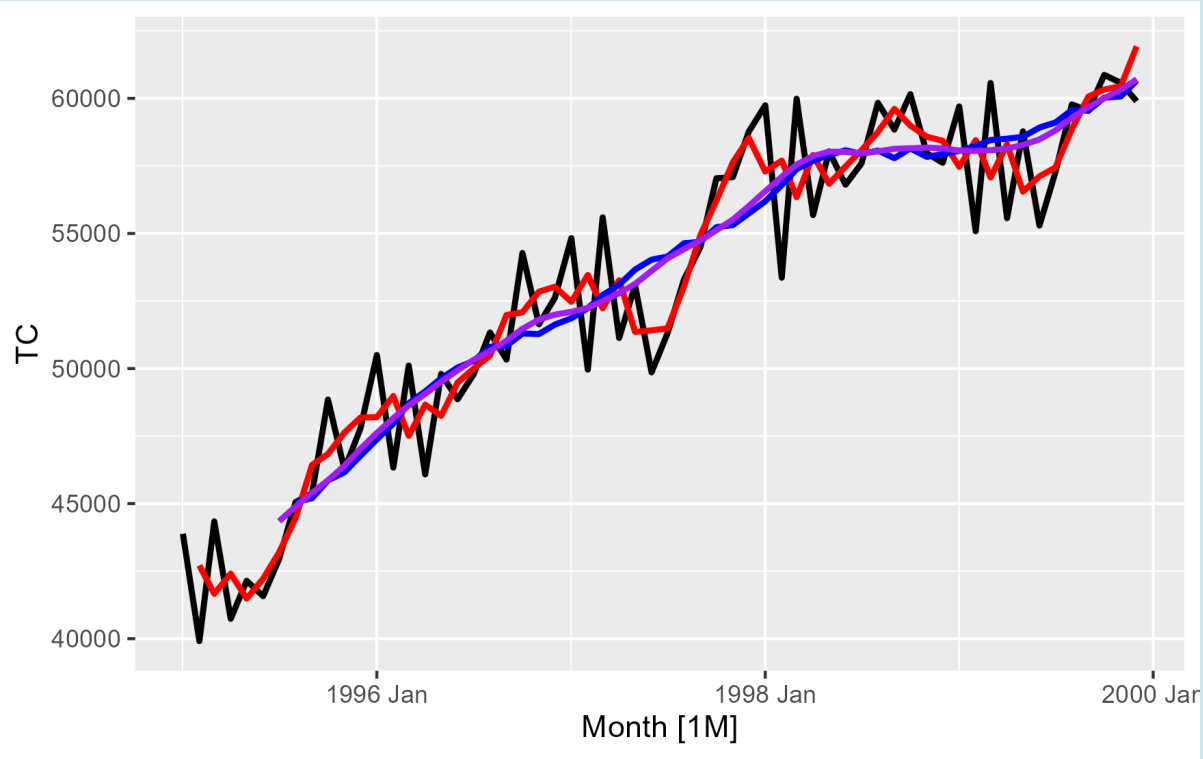
Question **14**

Correct

Mark 1.00 out of 1.00

Consider the following plot, where the black data represents the original data, while the 3 colored lines represent a 3-MA, 15-MA and 2x12-MA.

Which of the following statements are true?



- [ ] The 15-MA is the purple line.
- [x] The 15-MA is the blue line. ✓
- [x] The 2x12-MA is the purple line ✓
- [ ] The 3-MA is the blue line.
- [ ] The 3-MA is the purple line.
- [ ] The 2x12-MA is the red line.
- [ ] The 15-MA is the red line.
- [x] The 3-MA is the red line. ✓
- [ ] The 2x12-MA is the blue line.

Your answer is correct.

3-MA is the red one, since that is the most uneven.

The blue one, while a lot smoother then the red one, still has some zig-zags, so it is the 15-MA.

The purple one is the 2x12-MA, which eliminates any possible seasonal effect from the data.

The correct answers are: The 3-MA is the red line., The 15-MA is the blue line., The 2x12-MA is the purple line

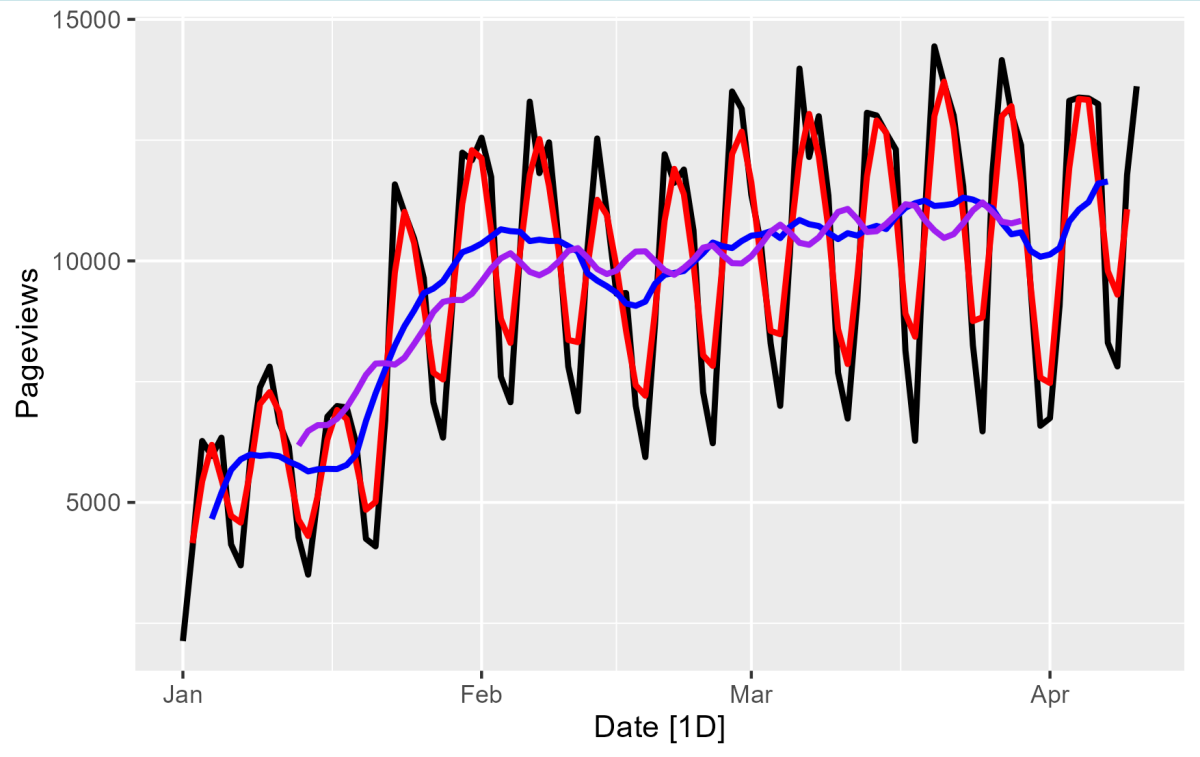
Question **15**

Correct

Mark 1.00 out of 1.00

Consider the following plot, where the black data represents the original data, while the 3 colored lines represent a 3-MA, 7-MA and 25-MA.

Which of the following statements are true?



- [x] The 25-MA is the purple line
- [ ] The 25-MA is the red line.
- [ ] The 25-MA is the blue line.
- [x] The 7-MA is the blue line.
- [ ] The 7-MA is the red line.
- [ ] The 7-MA is the purple line.
- [ ] The 3-MA is the blue line.
- [x] The 3-MA is the red line.
- [ ] The 3-MA is the purple line.

Your answer is correct.

3-MA is the red one, since that is the most uneven.

The blue one, since that is the smoothest is the 7-MA, since in case of daily data, that is the length of the seasonal period.

The purple one is the 25-MA. Quite smooth, but since the seasonal period has 7 observations, it is not as smooth as the blue one.

Obs. You can also look at how many observations are missing from the two ends of the plot. The shorter the line, the higher the order of the MA.

The purple one is the 2x12-MA, which eliminates any possible seasonal effect from the data.

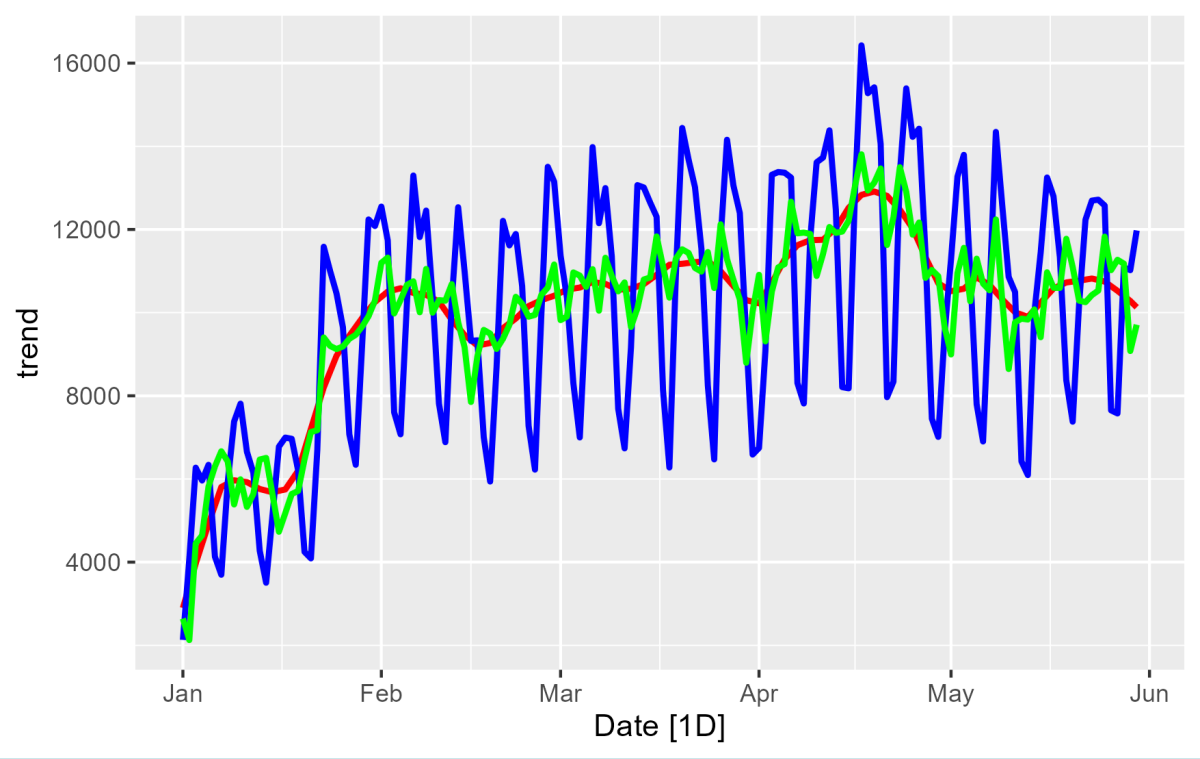
The correct answers are: The 3-MA is the red line., The 7-MA is the blue line., The 25-MA is the purple line

Question **16**

Correct

Mark 1.00 out of 1.00

On the plot below we have the original data, and the trend and seasonally adjusted component returned by an STL decomposition. Which of the following statements are true?



- [x] The trend is the red one.
- [ ] The trend is the green one.
- [ ] The original data is the green one.
- [ ] The seasonally adjusted component is the blue one.
- [x] The original data is the blue one.
- [ ] The trend is the blue one.
- [ ] The seasonally adjusted component is the red one.
- [x] The seasonally adjusted component is the green one.
- [ ] The original data is the red one.

Your answer is correct.

Trend should be smooth (so it is the red line). Seasonally adjusted component is trend + error, which is the green one. This leaves the blue as the original data.

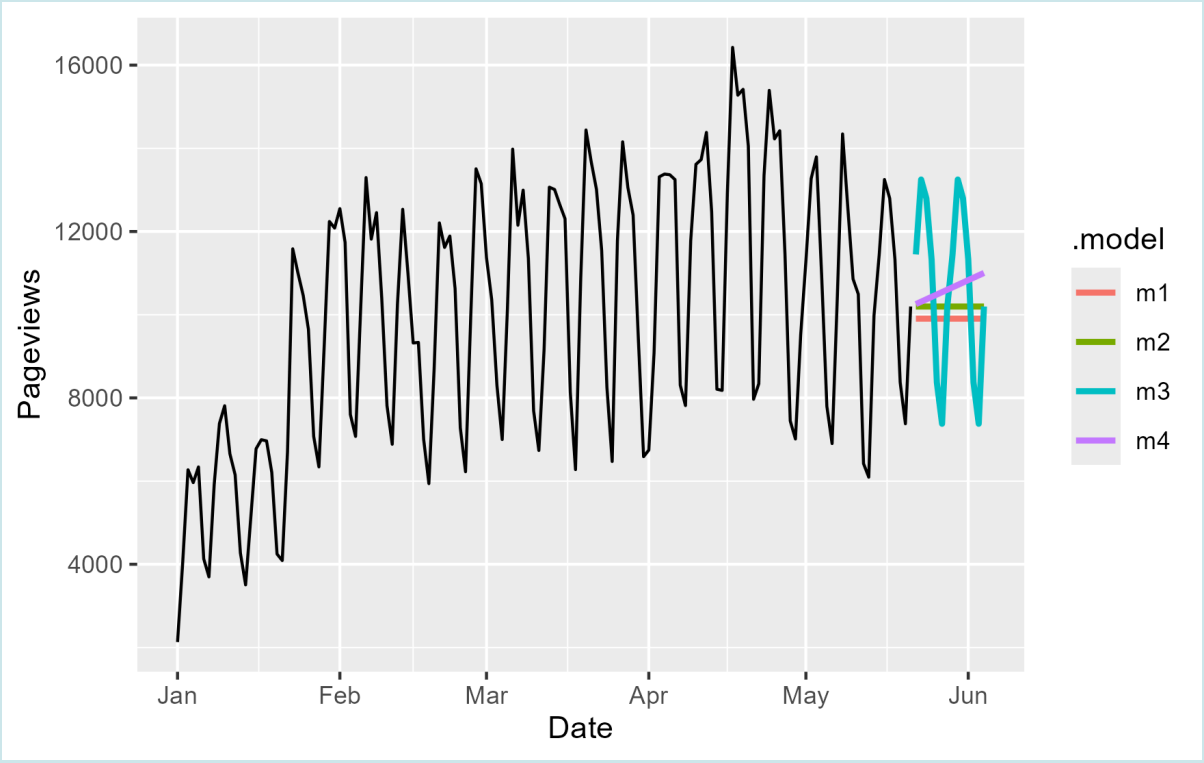
The correct answers are: The original data is the blue one., The trend is the red one., The seasonally adjusted component is the green one.

Question **17**

Correct

Mark 1.00 out of 1.00

On the plot below, we have the forecasts for 14 days for the 4 baseline methods. Match the plot color with the method that generated the forecast.



m2 (green)	Naive
m1 (red)	Mean
m3 (blue)	Seasonal naive
m4 (purple)	Drift

Your answer is correct.

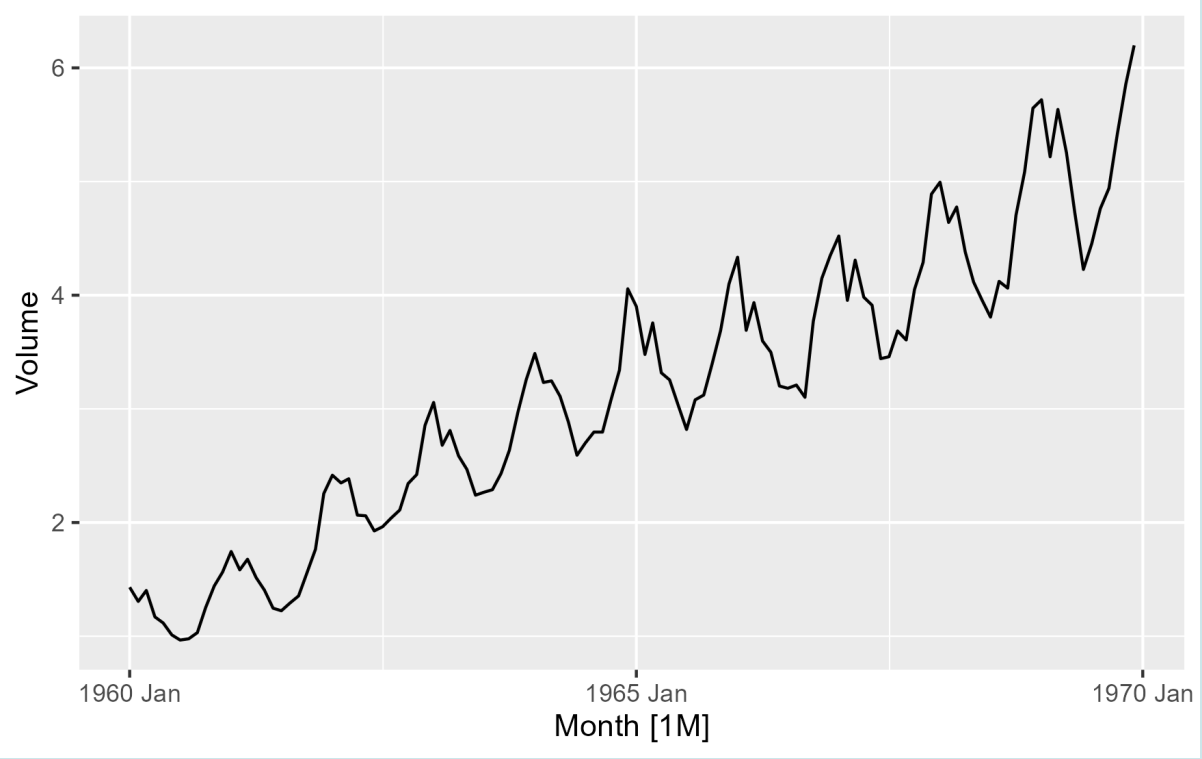
The correct answer is: m2 (green) → Naive, m1 (red) → Mean, m3 (blue) → Seasonal naive, m4 (purple) → Drift

Question **18**

Partially correct

Mark 0.50 out of 1.00

Consider the following time series plot. Which of the forecasting methods from below might be suitable for this data?



- [x] A regression model with 4 pairs of Fourier terms. ✓
- [x] A linear regression model based on dummies for the seasons. ✗
- [ ] An ETS(A, N, N) model
- [ ] An ETS(M, Ad, N) model
- [x] An ETS(A, Ad, A) model. ✗ The seasonal component seems to be multiplicative.
- [ ] An ARIMA(1, 0, 1)(1, 0, 1)_12 model.
- [ ] An ETS(A, Ad, M) model.
- [ ] Seasonal naive
- [x] A decomposition model, where we use drift for the seasonally adjusted component and seasonal naive for the seasonal component. ✓
- [x] An ARIMA(1,0,1)(2,1,2)_12 model. ✓
- [ ] An ARIMA(1,0,1)(2,1,3)_4 model.

- [ ] An ARIMA(1, 0, 1) model.

Your answer is partially correct.

You have selected too many options.
We have monthly data with a strong seasonality and trend. We need methods which have both trend and a seasonal component. Also, seasonality seems to be multiplicative.

The correct answers are: An ETS(A, Ad, M) model., An ARIMA(1,0,1)(2,1,2)_12 model., A decomposition model, where we use drift for the seasonally adjusted component and seasonal naive for the seasonal component., A regression model with 4 pairs of Fourier terms.

Question **19**

Incorrect

Mark 0.00 out of 1.00

For a given time series, we build the following models and get the following AIC_c values. Which model is the best?

ETS(A, N, N) - 278.37

ETS(M, Ad, N) - 256.33

ETS(A, Ad, A) - 177.23

ETS(A, Ad, M) - 111.87

ARIMA(1, 0, 1) - -116.16

ARIMA(1, 0, 1)(2, 1, 2)_12 - -114.82

TSLM with 4 pairs of Fourier terms - 68.6

- [ ] ETS(A, N, N)
- [x] ARIMA(1, 0, 1) ⊗
- [ ] ETS(A, Ad, A)
- [ ] ARIMA(1, 0, 1)(2, 1, 2)_12
- [ ] It cannot be determined.
- [ ] TSLM with 4 pairs of Fourier terms
- [ ] ETS(M, Ad, N)
- [ ] ETS(A, Ad, M)

Your answer is incorrect.

You cannot compare models produced by different methods based on AIC. You can compare the ETS methods (so ETS(A, Ad, M) is the best ETS), but cannot compare ETS with ARIMA for example. In case of ARIMA, differencing needs to be the same to be able to do a comparison.

The correct answer is: It cannot be determined.

Question **20**

Correct

Mark 1.00 out of 1.00

For a given time series with monthly data, we build the following models on 120 observations, then produce forecasts for the next year and compare the values of the forecasts to a test set of 12 observations, not used for training the models. We get the following MAPE values. Which model is the best?

Seasonal naive - 13.2

ETS(A, N, N) - 9.55

ETS(M, Ad, N) - 11.5

ETS(A, Ad, A) - 4.47

ETS(A, Ad, M) - 4.23

ARIMA(1, 0, 1) - 4.43

ARIMA(1, 0, 1)(2, 1, 2)_12 - 3.93

TSLM with 4 pairs of Fourier terms - 47.3

- [ ] ETS(A, Ad, M)
- [ ] ETS(A, Ad, A)
- [ ] ETS(A, N, N)
- [ ] ARIMA(1, 0, 1)
- [x] ARIMA(1, 0, 1)(2, 1, 2)_12
- [ ] ETS(M, Ad, N)
- [ ] It cannot be determined.
- [ ] TSLM with 4 pairs of Fourier terms

Your answer is correct.

This is the proper way of comparing models. The best is the one with the smallest MAPE.

The correct answer is: ARIMA(1, 0, 1)(2, 1, 2)_12

◄ Lecture 3 - 4 Quiz

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